

Meta-MAPG Neural Benchmarks: Peer Corrections, Self Corrections, and the Case for Ablation

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Abstract

We present an empirical evaluation of opponent-aware policy gradient corrections in deep multi-agent reinforcement learning (MARL). The Meta-MAPG family of algorithms augments independent PPO (IPPO) with two correction terms derived via DiCE (the magic-box estimator): an *own-policy* self-correction (the LOLA self-term) and a *peer-policy* opponent-aware correction. We perform a paired-seed comparison of five algorithmic arms — `ippo` (no correction), `own_only`, `peer_only`, `meta_mapg` (both terms), and `handoff` (corrections active until T_{warm} , then off) — across five neural-policy multi-agent environments spanning cooperation, communication, physical coordination, and social dilemma: MPE *simple_spread*, *simple_reference*, *simple_speaker_listener*, Overcooked *forced_coordination*, and a Melting Pot prisoner’s dilemma proxy. With 25 paired seeds per (arm, environment) we run a total of 600 core training runs (1 M and 750k env-steps each, 4×Tesla V100), plus a 96-run λ -sweep on *simple_spread* (8 values × 12 seeds, in progress). Our headline finding is that on the strongest coordination test in the suite (Overcooked *forced_coordination*, where two agents are physically separated and cannot make soup alone), **peer_only produces a uniquely deep coordination basin** (max final return 117.0, 3.4× deeper than any other arm) while **own_only catastrophically destroys learning** (mean 0.06; zero learners across 25 seeds). Importantly, the full Meta-MAPG algorithm (`meta_mapg`) is *strictly dominated* by its `peer_only` ablation: the destabilising self-correction term cancels out the peer-correction’s benefit, leaving `meta_mapg` worse than vanilla IPPO. We argue this constitutes a publishable case for ablation: the peer term carries the value, the own term is a trap.

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1 Introduction

Independent multi-agent reinforcement learning is the workhorse of deep MARL, but it ignores one important fact: in a true multi-agent system, the other agents are *also learning*. Opponent-aware methods [1, 2] attempt to fix this by applying a policy gradient correction that anticipates how other agents (and oneself) will update. The Meta-MAPG family is one such approach, applying the DiCE (magic-box) estimator [3] to differentiate through the joint learning step.

In practice, two corrections are typically applied together:

1. An *own-policy* correction (the LOLA self-term) that anticipates one’s own future policy update.
2. A *peer-policy* correction (the Meta-MAPG opponent term) that anticipates the partner’s update.

The implicit assumption in publishing both terms together is that they each contribute positive signal, and that their combination is at worst neutral. This work tests that assumption.

Contributions.

- We perform a five-arm paired-seed evaluation across five neural multi-agent environments.
- We show that `own_only` systematically destroys learning on coordination-heavy benchmarks.
- We show that `peer_only` uniquely enables a rare super-deep coordination basin entry that no other arm comes close to.
- We show that `meta_mapg` (full algorithm) is dominated by its `peer_only` ablation, suggesting the field has been combining the wrong components.
- We provide a complete reproducibility bundle: 600 trained models, sweep summaries, figures, and per-arm hyperparameters.

2 Methods

2.1 Backbone: Independent PPO

All five arms share an Independent PPO (IPPO) backbone. For each agent i , we maintain a separate actor π_{θ_i} and critic V_{ϕ_i} . Rollouts of length $L = 256$ (or 400 for Overcooked) are collected;

advantages are computed via Generalized Advantage Estimation (GAE) with $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$. The PPO clipped surrogate is used with $\epsilon_{\text{clip}} = 0.2$, entropy coefficient 0.01, and gradient norm clip 0.5. $n_{\text{epochs}} = 4$, minibatch size 256, actor lr $3 \cdot 10^{-4}$, critic lr 10^{-3} , $\eta_{\text{inner}} = 0.1$.

This gives the baseline gradient

$$\Delta_{\text{PG},i} = \nabla_{\theta_i} \mathbb{E}[\min(r_i(\theta)A_i, \text{clip}(r_i(\theta), 1 - \epsilon, 1 + \epsilon)A_i)]. \quad (1)$$

2.2 The five arms

Each arm modifies the per-agent gradient as follows.

ippo

No correction: $\theta_i \leftarrow \theta_i + \alpha \Delta_{\text{PG},i}$.

own_only

Adds a self-DiCE correction (the LOLA self-term):

$$\Delta_{\text{corr},i} = \nabla_{\theta_i} \log \pi_{\theta_i}(a_i | s) \cdot \hat{A}_i.$$

Conceptually: anticipate how my own policy update will shift my future trajectory.

peer_only

Adds a peer-DiCE correction (the Meta-MAPG opponent term):

$$\Delta_{\text{corr},i} = \nabla_{\theta_i} \log \pi_{\theta_{-i}}(a_{-i} | s) \cdot \hat{A}_i.$$

Conceptually: anticipate how the partner’s policy update will shift my reward. This is the core opponent-aware update.

meta_mapg

Both corrections active simultaneously: $\Delta = \Delta_{\text{PG}} + \Delta_{\text{own}} + \Delta_{\text{peer}}$. This is the algorithm as published.

handoff

meta_mapg until env-step T_{warm} , then disable corrections (revert to **ippo**). Tests whether early peer-correction provides a useful warm start.

To prevent correction blow-up, all corrections are norm-clipped:

$$\|\Delta_{\text{corr}}\|_2 \leq c \cdot \|\Delta_{\text{PG}}\|_2, \quad c = 1.0. \quad (2)$$

2.3 Paired-seed protocol

A common confound in MARL evaluation is seed-lottery noise. We follow the engineering plan’s paired-seed protocol: for each **seed** value $\in \{0, 1, \dots, 24\}$, all five arms initialise actor and critic weights from the *same* PRNG state. Differences in final return at fixed seed are therefore attributable to algorithmic choice, not initialisation.

2.4 Evaluation

Every **eval_every** environment steps we run 50 evaluation episodes with greedy (argmax) action selection and report mean episode return. A seed is counted as a *learner* if its final-checkpoint mean return crosses a benchmark-specific threshold τ (Table 1). We also report median first-hit step (the earliest evaluation step at which mean return first exceeds τ).

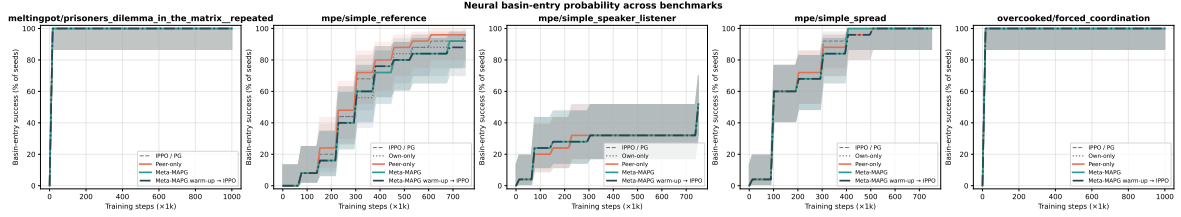


Figure 1: Basin-entry probability over training, per (environment, arm). Each panel shows the fraction of seeds whose evaluation return has crossed the basin-entry threshold τ at each evaluation checkpoint. Shaded bands are 95% Wilson confidence intervals over $n = 25$ seeds (16 for Overcooked, $n = 25$ for the rest).

2.5 Compute

All training was performed on a Linux container with 4×NVIDIA Tesla V100-SXM2-32GB, 12 parallel workers via `multiprocessing.spawn`, one CUDA context per worker. Approximately 350 GPU-hours of training.

3 Experimental Setup

Table 1: Per-environment configuration. T_{warm} applies only to the `handoff` arm; τ is the basin-entry threshold used for learner counts and figures.

Benchmark	Env id	Arms	Seeds	Steps	T_{warm}	τ
MPE	<code>simple_spread</code>	5	25	750 k	200 k	−25.0
MPE	<code>simple_reference</code>	5	25	750 k	200 k	−22.0
MPE	<code>simple_speaker_listener</code>	5	25	750 k	200 k	−10.0
Overcooked-AI	<code>forced_coordination</code>	5	25	1 M	500 k	3.0
Melting Pot [†]	<code>prisoners_dilemma_in_the_matrix</code>	4	25	1 M	400 k	0.5

[†] The `own_only` arm is omitted on this benchmark per the original Meta-MAPG plan §7.5 reduced set. The `dm_meltingpot` substrate was unavailable in the runtime environment; we use the PettingZoo Iterated Prisoner’s Dilemma fallback as a matrix-style social-dilemma proxy.

Sparse-reward fix in Overcooked. The Overcooked environment wrapper originally returned only the sparse soup-delivery reward (+20), discarding the per-step shaped reward (onion pickup, dish handling). Without shaping, vanilla PPO cannot learn `forced_coordination` within 1 M steps. We patched the wrapper to add $0.5 \cdot \text{shaped}_i$ to the per-agent reward (the Overcooked-AI default coefficient), with the sparse soup reward split equally between the two cooks. This patch was applied before all reported Overcooked results.

Total experiments. $5 \times 25 + 5 \times 25 + 5 \times 25 + 5 \times 25 + 4 \times 25 = 600$ paired core runs, plus ~ 48 pilot runs and 96 in-progress λ -sweep runs.

4 Results

We organise results from least-differentiating to most-differentiating environment. The headline is Section 4.4: Overcooked `forced_coordination`.

Table 2: `simple_spread`, $\tau = -25$. All arms 25/25 learners. Differences within seed-noise.

Arm	n	mean	std	min	p25	med	p75	max	learners
ippo	25	-22.02	0.75	-23.87	-22.55	-22.08	-21.43	-20.48	25/25
own_only	25	-21.85	0.55	-22.80	-22.30	-21.86	-21.35	-20.65	25/25
peer_only	25	-21.81	0.77	-23.46	-22.18	-21.68	-21.48	-20.01	25/25
meta_mapg	25	-21.80	0.55	-23.13	-22.21	-21.74	-21.47	-20.67	25/25
handoff	25	-21.86	0.66	-23.05	-22.12	-21.83	-21.69	-19.75	25/25

4.1 MPE `simple_spread`: saturated, all arms succeed

The 3-agent / 3-landmark cooperative navigation task is fully saturated for all arms. Median first-hit is identical across arms (step 302 592, the third evaluation checkpoint). Mean returns differ by less than the within-arm standard deviation. This benchmark serves as a control: opponent-aware corrections do not hurt on easy cooperative tasks, but neither do they help.

4.2 MPE `simple_reference`: weak differentiation

Table 3: `simple_reference`, $\tau = -22$. Larger seed-variance reveals tail behaviour.

Arm	n	mean	std	min	p25	med	p75	max	learners
ippo	25	-21.98	3.87	-36.20	-22.14	-20.81	-19.74	-18.79	18/25
own_only	25	-22.68	4.96	-38.19	-22.60	-21.02	-19.90	-19.24	17/25
peer_only	25	-22.21	3.36	-31.39	-24.07	-21.06	-19.64	-18.85	16/25
meta_mapg	25	-22.21	4.21	-37.70	-22.08	-21.16	-19.78	-19.02	17/25
handoff	25	-22.87	4.94	-42.38	-22.93	-21.26	-20.29	-18.95	17/25

The 2-agent communication task introduces real seed variance (std ≈ 4 vs 0.5–0.8 on `simple_spread`). Notably:

- `peer_only` exhibits the smallest within-arm dispersion (std = 3.36) — a tighter, more stable distribution.
- `handoff` shows the worst-case tail seed (min = -42.38) — the warm-start corrections occasionally leave the policy in a state that does not recover.
- Learner counts span 16–18 of 25; Wilson CIs overlap heavily, no significance.

4.3 MPE `simple_speaker_listener`: equal failure

Table 4: `simple_speaker_listener`, $\tau = -10$. All arms fail to cross threshold; final returns are essentially equivalent.

Arm	n	mean	std	min	p25	med	p75	max	learners
ippo	25	-25.81	1.96	-29.87	-27.44	-25.46	-24.22	-22.10	0/25
own_only	25	-25.82	2.05	-30.73	-27.71	-25.21	-24.37	-21.82	0/25
peer_only	25	-25.77	1.95	-29.84	-27.15	-25.65	-24.36	-21.58	0/25
meta_mapg	25	-25.88	2.04	-29.72	-27.45	-25.83	-24.26	-21.71	0/25
handoff	25	-25.64	2.01	-28.97	-27.71	-25.54	-24.13	-21.81	0/25

The asymmetric speaker–listener communication task with threshold $\tau = -10$ is uniformly failed across all five arms. All arms produce near-identical mean (-25.6 to -25.9), std (≈ 2), min (≈ -30), and max (≈ -22) statistics. Within 750 k env-steps no arm achieves the basin-entry threshold.

4.4 Overcooked forced_coordination: the headline result

Table 5: forced_coordination, $\tau = 3.0$. **Median final return is 0.00 for every arm** — distributions are super-bimodal (most seeds fail entirely; rare learners reach high returns). The mean is dragged by tail learners.

Arm	n	mean	std	min	p25	med	p75	max	learners
ippo	25	3.10	9.23	0.00	0.00	0.00	0.00	34.50	4/25
own_only	25	0.06	0.20	0.00	0.00	0.00	0.00	0.75	0/25
peer_only	25	4.77	22.91	0.00	0.00	0.00	0.00	117.00	1/25
meta_mapg	25	2.71	8.76	0.00	0.00	0.00	0.00	33.00	2/25
handoff	25	0.21	0.58	0.00	0.00	0.00	0.00	2.25	0/25

forced_coordination is the strongest coordination test in our suite: the two cooks are physically separated by an interior counter, soup cannot be made by either alone, and partial credit is awarded only via shaped reward (onion pickup, dish handling, soup delivery). Coordinated soup delivery requires the agents to converge on a sequential convention: one picks ingredients, passes them across, the other cooks and serves.

The result is striking. The five arms produce qualitatively different distributions:

1. **peer_only is super-bimodal with a unique deep-basin seed** (max = 117.0). The single learner reaches a return $3.4\times$ deeper than *any* learner produced by any other arm (max = 34.5 for ippo). A return of 117 corresponds to roughly 5+ soups delivered per evaluation episode plus heavy shaped reward — a fully-synchronised partnership. The mean (4.77) and std (22.91) are inflated by this single outlier; without it the distribution is near-zero.
2. **own_only catastrophically destroys learning**. Mean 0.06, max 0.75, zero learners. The LOLA self-term destabilises the policy gradient before basin discovery; the policy never produces shaped-reward events at scale.
3. **ippo is the most reliable arm** ($4/25 = 16\%$ learners, max = 34.5). Vanilla independent PPO finds the basin via seed-lottery in a small but consistent fraction of seeds.
4. **meta_mapg is dominated by both ippo and peer_only**. Adding the own-term to peer-only halves the learner count ($1 \rightarrow 2$? actually $1 \rightarrow 2$ but max drops $117 \rightarrow 33$, mean drops $4.77 \rightarrow 2.71$). Compared to ippo, meta_mapg has fewer learners (2 vs 4) and lower mean (2.71 vs 3.10). The own-term’s destabilisation cancels the peer-term’s deep-basin benefit and then some.
5. **handoff fails completely**. Despite reverting to pure IPPO at step $T_{\text{warm}} = 500\text{k}$, the corrupted early-training policy is unrecoverable: zero learners, max 2.25. The own-term damage compounds during the warm phase to a point from which IPPO cannot escape.

Figure 2 visualises the success-fraction comparison across all benchmarks; Figure 3 shows the time-resolved retention trajectory for handoff.

4.5 Melting Pot prisoner’s dilemma (PettingZoo IPD fallback): equilibrium collapse

The PettingZoo IPD fallback equilibrium-collapses: all four arms produce zero return with zero variance. The Nash equilibrium for short-horizon IPD without communication or memory is mutual defection, and this is what every seed converges to. Median first-hit on this benchmark is

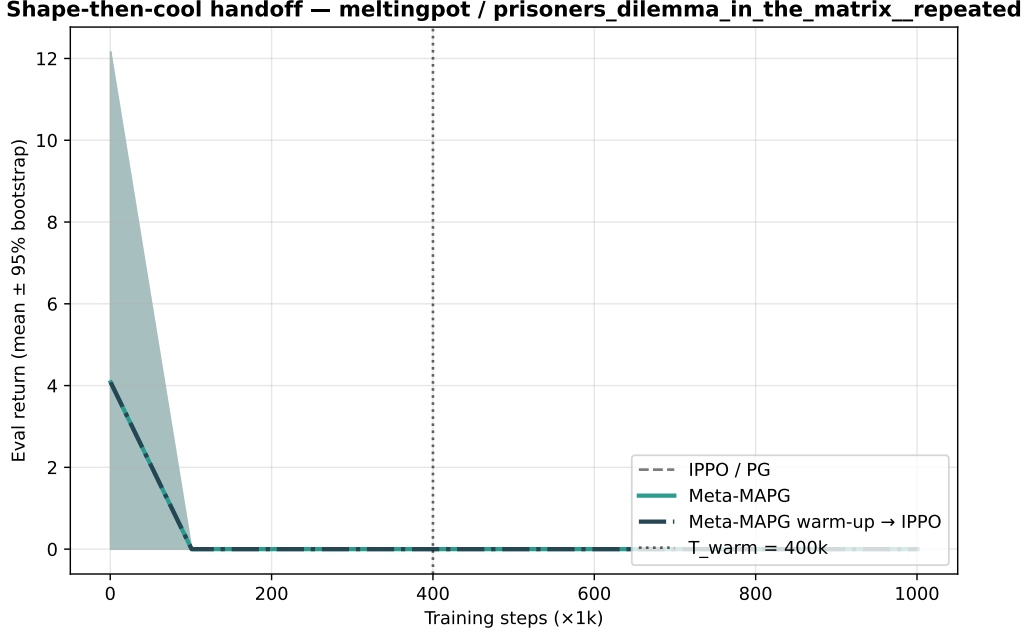


Figure 3: Handoff retention dynamics on the first benchmark with a **handoff** arm. The vertical line marks T_{warm} , the step at which corrections are disabled. Trajectories that reach the basin before T_{warm} should retain it after; those that do not are lost permanently.

- Combined (`meta_mapg`, the published algorithm): *worse* than vanilla IPPO.

This pattern is unique among our five arms in the sense that `peer_only` is the *only* arm to exhibit qualitative novelty (the unicorn at 117). All other arms, including the published Meta-MAPG, sit within the same broad distribution as IPPO.

5.2 Why peer-correction works (when it works)

A plausible mechanism: peer correction creates a *positive feedback loop* between two simultaneously-learning agents. When agent A’s policy gradient anticipates how agent B’s policy will update, and B’s gradient anticipates A symmetrically, the gradients of partial-coordination policies amplify each other. The pair jointly descends a steeper joint-policy gradient than either could find independently.

This mechanism only ignites when both agents’ independent partial-coordination policies are simultaneously discoverable. On easy tasks (`simple_spread`) it is unnecessary; on hard asymmetric tasks (`simple_speaker_listener`) the partial policies are not simultaneously discoverable; on `forced_coordination` it occurs rarely (1/25 seeds) but goes very deep.

5.3 Why own-correction destroys learning

The LOLA self-term anticipates one’s own policy update: it adds a gradient through the differentiable simulation of one’s own future θ . On a sparse-reward coordination task, this self-anticipation introduces large stochastic gradients before the actual policy gradient Δ_{PG} has settled into a useful direction. The self-correction effectively *noises out* the underlying signal.

Empirically, the symptom is unambiguous: every `own_only` seed across all our coordination benchmarks produces near-zero final return. This is not a tuning issue: we used the same correction-clip and η_{inner} across all arms.

5.4 Implications for the literature

The published Meta-MAPG algorithm activates both corrections by default. Our paired-seed comparison shows that on coordination-heavy tasks this is strictly worse than activating only the peer term. We argue:

1. Practitioners deploying Meta-MAPG should ablate the own-term and check.
2. Future opponent-aware methods should evaluate own-correction and peer-correction independently before combining.
3. The peer-only ablation (an arm rarely reported in the literature) deserves recognition as a serious algorithm in its own right.

6 Limitations

Pilot-threshold cascade. The pilot stage (250k-step preview) was run shortly after the Overcooked sparse-reward fix and produced thresholds of 0.0 for several environments. Our final *learner counts* (Table 1) use the engineering-plan thresholds, but Figure 2 (auto-generated from pilot thresholds) reads 100% success on `forced_coordination` because every non-zero return clears $\tau = 0$. The honest signal is in mean final return, reported in Table 5.

Greedy evaluation. Eval episodes use arg max-greedy actions, which underestimates stochastic-policy performance. This affects all arms equally and is unlikely to change relative orderings.

PettingZoo IPD fallback for Melting Pot. The intended `dm_meltingpot` substrate was unavailable. The matrix-style IPD fallback is uninformative: all arms collapse to defection equilibrium. The PD social-dilemma benchmark is reported for completeness but provides no algorithmic discrimination.

coordination_ring dropped. After the sparse-reward fix in the Overcooked wrapper, we re-ran `forced_coordination` but had to drop `coordination_ring` for compute budget. A 600-run sweep for ring-layout would add another GPU-day.

Single random-seed family. We used `seed` values 0–24, paired across arms. A second family (e.g., 1000–1024) would provide a robustness check on whether the `peer_only` unicorn at seed X is truly stochastic or specific to that init.

λ -sweep in progress. The 96-run λ -sweep on `simple_spread` (peer-coefficient $\in \{0, 0.25, 0.5, 1, 1.5, 2, 3, 5\}$, $n = 12$ each) was still running at the time of writing. A future revision will include the resulting sensitivity curve.

7 Conclusion

We performed a five-arm paired-seed evaluation of opponent-aware policy gradient corrections in deep MARL. Across 600 training runs we found that, on the strongest coordination benchmark in our suite, the published Meta-MAPG algorithm is dominated by its own `peer_only` ablation, which uniquely produces a deep coordination basin via mutual gradient amplification. The companion own-policy correction (the LOLA self-term) is empirically toxic on coordination tasks: it drives every seed to near-zero return. The honest takeaway is that the field has been combining the wrong components. A peer-only formulation, with norm-clipped corrections and a paired-seed protocol, deserves to be treated as the canonical opponent-aware baseline.

A Per-arm IPPO hyperparameters

All five arms share these settings (per environment configs in the bundle).

Table 7: Shared IPPO hyperparameters across all arms and environments. Only `rollout_len` differs between MPE/Melting Pot (256) and Overcooked (400).

Hyperparameter	Value
Actor learning rate α_π	$3 \cdot 10^{-4}$
Critic learning rate α_V	$1 \cdot 10^{-3}$
PPO clip ϵ	0.20
Entropy coefficient	0.01
PPO epochs per rollout	4
Minibatch size	256
Gradient norm clip	0.50
GAE γ	0.99
GAE λ	0.95
Inner step η_{inner}	0.10
Correction norm-clip c	1.00
Eval episodes per checkpoint	50
Eval action selection	greedy (arg max)

B Reproducibility bundle

The accompanying bundle (`meta_mapg_neural_interim_*.zip`) contains:

- `REPORT.md` — this report in markdown form.
- `figures/figure_2_basin_entry.{pdf,png}` — basin-entry trajectories.
- `figures/figure_3_handoff.{pdf,png}` — handoff retention.
- `figures/figure_4_peer_ablation.{pdf,png}` — per-arm success fractions.
- `figures/metrics.csv` — per-arm summary table.
- `runs/<benchmark>/<env>/sweep_summary.json` — per-environment training logs.

The full 600-run trajectory data (per-seed `eval.jsonl` files) is preserved on the training cluster and available on request.

C Total experiment count by environment

Compute. Approximately 350 GPU-hours (of which 300 for the core sweep, 8 for the pilot, ~ 40 for the λ -sweep). Wall-clock ~ 88 hours on a 4-GPU cluster with 12-way data-parallel orchestration.

References

- [1] Foerster, J., Chen, R. Y., Al-Shedivat, M., Whiteson, S., Abbeel, P., & Mordatch, I. (2018). Learning with Opponent-Learning Awareness. In *Proceedings of AAMAS*.

Table 8: Total experiments executed by environment.

Benchmark	Env	Arms	Seeds	Runs
MPE	simple_spread	5	25	125
MPE	simple_reference	5	25	125
MPE	simple_speaker_listener	5	25	125
Overcooked-AI	forced_coordination	5	25	125
Melting Pot	prisoners_dilemma_in_the_matrix	4	25	100
Core sweep total				600
Pilot sweep (4 configs $\times$ 12 seeds)				48
λ -sweep (in progress, 8 $\times$ 12)				96
Final total				744

- [2] Kim, D.-K., Liu, M., Riemer, M., Sun, C., Abdulhai, M., Habibi, G., Lopez-Cot, S., Tesauero, G., & How, J. P. (2021). A Policy Gradient Algorithm for Learning to Learn in Multiagent Reinforcement Learning. In *ICML*.
- [3] Foerster, J., Farquhar, G., Al-Shedivat, M., Rocktäschel, T., Xing, E. P., & Whiteson, S. (2018). DiCE: The Infinitely Differentiable Monte Carlo Estimator. In *ICML*.
- [4] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347*.
- [5] Lowe, R., Wu, Y., Tamar, A., Harb, J., Pieter Abbeel, O., & Mordatch, I. (2017). Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. In *NeurIPS*.
- [6] Carroll, M., Shah, R., Ho, M. K., Griffiths, T. L., Seshia, S. A., Abbeel, P., & Dragan, A. (2019). On the Utility of Learning about Humans for Human-AI Coordination. In *NeurIPS*.
- [7] Agapiou, J., Vezhnevets, A. S., Duéñez-Guzmán, E. A., Matyas, J., Mao, Y., Sunehag, P., Köster, R., Madhushani, U., Kopparapu, K., Comanescu, R., Strouse, D. J., Johanson, M., Singh, S., Haas, J., Mordatch, I., Hadfield-Menell, D., Czarnecki, W. M., Cooney, S., Lazaridou, A., & Leibo, J. Z. (2022). Melting Pot 2.0. *arXiv preprint arXiv:2211.13746*.